

Assignment 2

Call or Fold?

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Question 1

Given,

$$X_1 \sim \text{Poisson}(\lambda), \quad X_2|X_1 = x_1 \sim \text{Bin}(x_1, p).$$

(i. Marginal PMFs

For X_1 , over x_1

$$f_{X_1}(x_1) = e^{-\lambda} \frac{\lambda^{x_1}}{x_1!}, \quad x_1 = 0, 1, 2, \dots$$

Hence,

$$X_1 \sim \text{Poisson}(\lambda).$$

For X_2 , let $x_1 = x_2 + k$ for some $+ve$ k . Then,

$$f_{X_2}(x_2) = \sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^{x_2+k}}{(x_2+k)!} \binom{x_2+k}{x_2} p^{x_2} (1-p)^k.$$

Simplifying,

$$f_{X_2}(x_2) = e^{-\lambda p} \frac{(\lambda p)^{x_2}}{x_2!},$$

so

$$X_2 \sim \text{Poisson}(\lambda p).$$

ii. Independence of X_2 and $Y = X_1 - X_2$

Let

$$Y = X_1 - X_2.$$

Replacing X_1 by $X_2 + Y$ in the joint PMF,

$$f_{X_2, Y}(x_2, y) = \frac{e^{-\lambda p} (\lambda p)^{x_2}}{x_2!} \cdot \frac{e^{-\lambda(1-p)} (\lambda(1-p))^y}{y!}.$$

WKT, this is the product of two marginal PMFs,

$$f_{X_2, Y}(x_2, y) = f_{X_2}(x_2) f_Y(y),$$

it follows that

$$X_2 \perp Y,$$

we get

$$Y \sim \text{Poisson}(\lambda(1-p)).$$

Question 2

Each box contains n coins, of which m are counterfeit. Therefore,

$$P(\text{false coin}) = \frac{m}{n}.$$

Since one coin is selected independently from each of the n boxes,

$$R \sim \text{Binomial}\left(n, \frac{m}{n}\right).$$

Hence,

$$P(R = r) = \binom{n}{r} \left(\frac{m}{n}\right)^r \left(1 - \frac{m}{n}\right)^{n-r}, \quad r = 0, 1, \dots, n.$$

As $n \rightarrow \infty$ with m fixed,

$$R \Longrightarrow \text{Poisson}(m),$$

whose PMF is

$$P(R = r) = e^{-m} \frac{m^r}{r!}.$$

Question 3

i. Normalizing Constant

A Pdf satisfies

$$\int_0^\infty \int_0^{x_1} c e^{-x_1} dx_2 dx_1 = 1.$$

Evaluating it gives

$$c \int_0^\infty x_1 e^{-x_1} dx_1 = 1.$$

Since

$$\int_0^\infty x e^{-x} dx = 1,$$

we get

$$\boxed{c = 1.}$$

ii. Conditional PDF

The marginal density of X_1 is

$$f_{X_1}(x_1) = \int_0^{x_1} e^{-x_1} dx_2 = x_1 e^{-x_1}.$$

Thus,

$$f_{X_2|X_1}(x_2|x_1) = \frac{f(x_1, x_2)}{f_{X_1}(x_1)} = \frac{1}{x_1},$$

for

$$0 \leq x_2 \leq x_1.$$

Therefore,

$$X_2|X_1 = x_1 \sim \text{Uniform}(0, x_1).$$

iii.

When $X_1 = 2$,

$$P(X_2 > 1|X_1 = 2) = \int_1^2 \frac{1}{2} dx = \frac{1}{2}.$$

Question 4

Let

$$S = \sum_{i=1}^N X_i,$$

where

$$N \sim \text{Poisson}(\lambda).$$

By using conditional expectation,

$$E[S] = E[E(S|N)] = E[N\mu] = \lambda\mu.$$

Applying the law of total variance

$$\begin{aligned}\text{Var}(S) &= E[\text{Var}(S|N)] + \text{Var}(E[S|N]) \\ &= E[N\sigma^2] + \text{Var}(N\mu) \\ &= \lambda\sigma^2 + \lambda\mu^2.\end{aligned}$$

Hence,

$$\boxed{\text{Var}(S) = \lambda(\sigma^2 + \mu^2).}$$

Question 5

i.

Let

$$U = X_1 + X_2, \quad V = X_1 - X_2.$$

Then

$$\begin{aligned}\text{Cov}(U, V) &= \text{Cov}(X_1 + X_2, X_1 - X_2) \\ &= \text{Var}(X_1) - \text{Var}(X_2).\end{aligned}$$

since

$$\text{Var}(X_1) = \text{Var}(X_2),$$

we obtain

$$\boxed{\text{Cov}(U, V) = 0.}$$

Hence proved U and V are uncorrelated.

ii.

Uncorrelated variables need not be independent.

Let

$$X \sim U(-1, 1), \quad Y = X^2.$$

Then we get

$$\text{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0,$$

because dist. is symmetric

But Y is completely determined by X , so the variables are dependent.

Hence,

Uncorrelatedness does not imply independence.
